

Frequency

A frequency counts how often a value or category occurs.

In 0, 1, 1, 2, 2, 2, 3, 4, what is the frequency of 2?

Frequency tables

Number of downloads for 12 sessions:

0, 1, 2, 1, 3, 2, 0, 2, 1, 4, 2, 2

What frequency table summarizes these data?

Relative frequency

Relative frequency = frequency / total number of observations.

If 5 of 12 sessions have two downloads, what is their relative frequency?

Percentages

Seven of 20 printers are in use.

What percentage of the printers is in use?

Cumulative relative frequency

Cumulative relative frequency adds relative frequencies through a value.

If frequencies for 0, 1, 2, 3 downloads are 2, 3, 4, 1, what fraction have at most 2?

Cumulative frequency

Cumulative frequency counts observations at or below a cutoff. Synthetic frequencies for 0, 1, 2, 3 downloads: 2, 3, 4, 1.

How many sessions have at most 2 downloads?

At most or less than?

Downloads: 0, 1, 1, 2, 2, 2, 3, 4

What percentage have fewer than 2 downloads?

More than a cutoff

Downloads: 0, 1, 1, 2, 2, 2, 3, 4

What percentage have more than 2 downloads?

Checking a table

A frequency table for 20 observations lists counts 4, 7, and 8.

What should we check before interpreting it?

Grouped data

Download times in seconds: 2, 4, 5, 7, 8, 9, 11, 14

**How many times belong in each interval:
0 to less than 5, 5 to less than 10, 10 to
less than 15?**

Reading a grouped frequency table

Synthetic duration classes $[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$ seconds have frequencies 2, 5, 8, 5.

What percentage of durations are at least 5 but less than 15 seconds?

Recovering interval frequency

Synthetic cumulative relative frequencies at upper boundaries 5, 10, 15, 20 seconds are 0.10, 0.35, 0.75, 1.00.

What relative frequency lies from 10 up to 15 seconds?

Rounding

Textbook convention: keep one more decimal place than the original data and round only the final result.

For synthetic values 4, 5, and 8, what mean should we report?

Rounding totals

Three equally frequent categories each have relative frequency $1/3$.

Why might displayed percentages fail to total exactly 100%?

Levels of measurement

A level of measurement describes which comparisons are meaningful: category, order, differences, or ratios.

Why should the measurement level affect which summaries we calculate?

Nominal level

The nominal scale has categories with no inherent order.

What is the level of measurement for a printer's brand?

Ordinal level

The ordinal scale orders categories without defined numerical gaps.

What can “poor, fair, good, excellent” tell us about service?

Interval level

The interval scale has meaningful numerical differences but no absolute zero representing none of the quantity.

Does 20 degrees Celsius represent twice the temperature of 10 degrees Celsius?

Ratio level

The ratio scale has meaningful differences and ratios, with a zero representing none of the measured quantity.

Does a 20-minute download take twice as long as a 10-minute download?

Measurement level

A tournament records first, second, and third place.

What level of measurement is this?

Measurement level

A file list records sizes in bytes.

What level of measurement is this?

Choosing a summary

A survey codes favorite browsers as 1, 2, 3, and 4.

Would the mean of these codes answer a useful question?